

Labwork 2: Get to Know Your GPU

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1 Objective

For this labwork, the main goal is to use numba.cuda to get to know my GPU

- Devicename
- Coreinfo:multiprocessorcount,corecount
- Memoryinfo:memorysize

2 Python code

```
from numba import cuda
print(cuda.detect())

gpu = cuda.select_device(0)
print("GPU name:", gpu.name)

print("Multiprocessor count:", gpu.MULTIPROCESSOR_COUNT)
print("Compute Capability:", gpu.compute_capability)
if gpu.compute_capability == (8,6):
    print("Cuda core count:", gpu.MULTIPROCESSOR_COUNT * 128)

free_memory, total_memory = cuda.current_context().get_memory_info()
print(f"Total GPU memory: {total_memory / (1024 ** 3):.2f} GiB")
print(f"Free GPU memory: {free_memory / (1024 ** 3):.2f} GiB")
```

Figure 1: Code

3 Results

```
(karma) PS C:\Users\lapla\Desktop\codespace\master 2\Distributed system\advancedhpc2026\lab2> python
.\lab2.py
Found 1 CUDA devices
id 0      b'NVIDIA GeForce RTX 3050 Ti Laptop GPU'      [SUPPORTED]
          Compute Capability: 8.6
          PCI Device ID: 0
          PCI Bus ID: 1
          UUID: GPU-bccf1f6c-0e58-117b-988a-c927c644ed55
          Watchdog: Enabled
          Compute Mode: WDDM
          FP32/FP64 Performance Ratio: 64
Summary:
      1/1 devices are supported
True
GPU name: b'NVIDIA GeForce RTX 3050 Ti Laptop GPU'
Multiprocessor count: 20
Compute Capability: (8, 6)
Cuda core count: 2560
Total GPU memory: 4.00 GiB
Free GPU memory: 3.22 GiB
```